

# Vishal Sivakumar

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## EDUCATION

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University of Alberta | Edmonton, AB

Expected Graduation: 17th June 2026

**B.S. Honors in Computer Science | GPA: 3.7/4.0 | Dean's Honor Roll (2022 - 2026)**

- Courses: Software Process & Product Management, Web Applications & Architecture, Operating Systems, File & Database Management, Data Structures and Algorithms

## TECHNICAL SKILLS

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**Programming Languages:** Java, Python, JavaScript, TypeScript, C, C++, C#, SQL

**Frameworks & Web:** Next.js, Node.js, Django, REST APIs, HTML/CSS, Android SDK, Express.js

**Databases & BaaS:** PostgreSQL, Supabase (PostgreSQL), Firebase, Relational Database Design

**Cloud & DevOps:** AWS (S3), Docker, Git, GitHub Actions CI/CD, Linux, Systems Development

**Methodologies & Tools:** SDLC, Agile/Scrum, GenAI Productivity Tools, Object-Oriented Design (OOD), JUnit, Espresso

## EXPERIENCE

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Stewards of Alberta's Protected Areas Association | Edmonton, AB

Jan. 2026 - Present

Software Process & Product Management | *CMPUT 401*

- Modernized legacy inspection workflows by delivering a full-stack platform (**Next.js, Node.js, TypeScript**), migrating 40,000+ individual system records into a high-availability production cloud layer.
- Optimized **Supabase (PostgreSQL)** schemas, relational queries, and backend logic to ensure high-performance data retrieval and long-term system scalability.
- Leveraged **Generative AI tooling** to accelerate rapid prototyping, automate comprehensive unit test drafting, and write technical documentation, significantly increasing development velocity.

Department of Computer Science | Edmonton, AB

Jan. 2026 - May. 2026

Undergraduate Teaching Assistant | *CMPUT 204 (Algorithms)*

- Selected as an **Algorithms Teaching Assistant** to mentor 400+ STEM students, clarifying complex problem-solving strategies, runtime algorithmic complexity (**O(n) analysis**), and core Data Structure implementations.
- Evaluated 200+ theoretical exams and assignments, rigorously grading algorithmic logic, optimization proofs, and structural problem-solving methodologies while supporting students via discussion forums.

## PROJECTS

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**Social Distribution** | Heroku, Python, Django Framework, PostgreSQL

Sep. 2025 - Dec. 2025

- **Collaborated** with a 6-member Agile team to develop a full-stack social web app using Python (Django), PostgreSQL on Heroku, and HTML/CSS/JavaScript frontend.
- **Engineered 30+** REST API endpoints to facilitate cross-platform integration, supporting a potential load of **100+** concurrent users.
- **Implemented** Heroku logging and basic monitoring for production services.

**Android Events Application** | Android Studio, Java, Firebase, Git

Sep. 2024 - Dec. 2024

- **Collaborated within a 6-member Agile team** utilizing Scrum practices to deliver a high-performance mobile application, ensuring rapid feature iteration and transparent sprint progress.
- **Integrated Firebase core services** to architect secure user authentication modules and handle real-time cloud data synchronization, optimizing client-to-server latency.
- **Enforced code reliability and long-term maintainability** by designing and executing comprehensive JUnit and Espresso UI testing suites, reducing post-deployment logic defects.